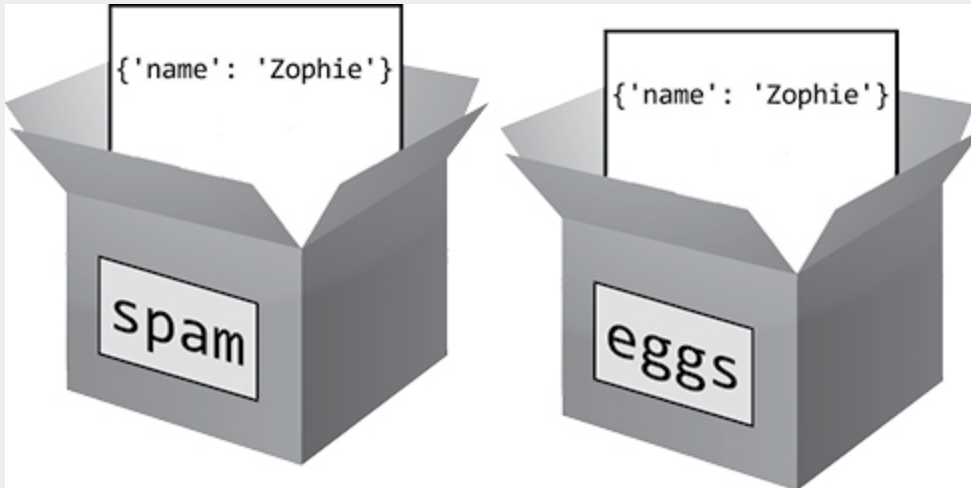


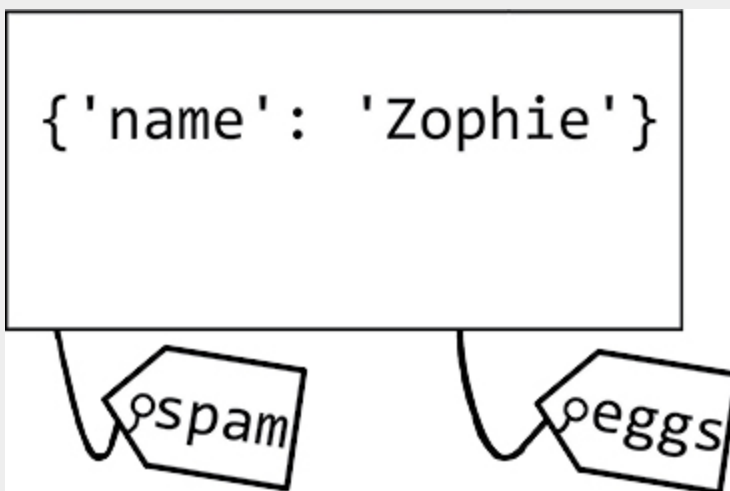
## VARIABLE METAPHORS: BOX VS. LABEL

Many introductory books use boxes as a metaphor for variables, which is an oversimplification. It's easy to think of variables as a box that a value is stored in, as in [Figure 7-1](#), but this metaphor falls apart when it comes to references. The previous spam and eggs variables don't store separate dictionaries; rather, they store *references* to the same dictionary in the computer's memory.



[Figure 7-1](#): Many books say you can think of a variable as a box that contains a value.

In Python, all variables are technically references, not containers of values, regardless of their data type. The box metaphor is simple but also flawed. Instead of thinking of variables as boxes, you can think of variables as labels for objects in memory. [Figure 7-2](#) shows labels on the previous spam and eggs examples.



[Figure 7-2](#): Variables can also be thought of as labels on values.